NOTATION · CHEAT SHEET

Trigonometry Notation at a Glance

The marks from the notation cards across the trigonometry pages: how each is read and what it means. Every group links to the full card on its page.

DEGREES AND RADIANS

$\theta,\ \alpha,\ \beta$	**THETA; ALPHA, BETA** Greek letters for angles; α, β when there are two.
$41^\circ 24' 36''$	**DEGREES, MINUTES, SECONDS** Base 60: $1^\circ = 60'$, $1' = 60''$.
$\sin 2$ vs $\sin 2^\circ$	**RADIANS UNLESS MARKED** A bare number is in radians.

RIGHT TRIANGLE

$\text{opp}, \text{adj}, \text{hyp}$	**OPPOSITE, ADJACENT, HYPOTENUSE** Opposite and adjacent depend on which angle you chose.
$\sin,\ \cos,\ \tan$	**SINE, COSINE, TANGENT** Function names: $\sin\theta$ is not $s \cdot i \cdot n \cdot \theta$.
$\csc,\ \sec,\ \cot$	**COSECANT, SECANT, COTANGENT** Reciprocals, crosswise: $\csc = \frac{1}{\sin}$, $\sec = \frac{1}{\cos}$.

UNIT CIRCLE

$(\cos\theta, \sin\theta)$	**THE POINT ON THE CIRCLE** Its coordinates are the cosine and sine.
I, II, III, IV	**QUADRANTS ONE TO FOUR** Counterclockwise from the upper right.
$\theta + 2\pi$	**THETA PLUS A FULL TURN** Same point, new label; also $\theta + 360^\circ$.

TRIGONOMETRIC FUNCTIONS

$\sin^2\theta$	**SINE SQUARED THETA** Means $(\sin\theta)^2$: square the value, not the angle.
$n \in \mathbb{Z}$	**N AN INTEGER** The formula holds for every integer n.
$n\pi,\ \frac{\pi}{2} + n\pi$	**MULTIPLES OF PI** Where $\sin = 0$, and where $\cos = 0$.

GRAPHS

$y = \sin x$	**Y EQUALS SINE X** On a graph the angle becomes the input x.
$A,\ B,\ C,\ D$	**THE FOUR PARAMETERS** In $y = A\sin(Bx - C) + D$: amplitude, frequency, shift, midline.
$T = \frac{2\pi}{\lvert B\rvert}$	**CAPITAL T, THE PERIOD** The shortest shift that repeats the graph.

INVERSE FUNCTIONS

$\sin^{-1}(x)$	**INVERSE SINE OF X** The inverse function, not $\frac{1}{\sin x}$.
$\arcsin,\ \arccos$	**ARCSINE, ARCCOSINE** The same functions, with no clash with powers.
`asin` , `atan2`	**NAMES IN CODE** Programming languages; `atan2` takes (y, x).

TRIGONOMETRIC EQUATIONS

$\frac{\pi}{6} + 2n\pi$	**A FAMILY OF SOLUTIONS** One answer for each integer n.
$(-1)^n \arcsin k$	**ALL SOLUTIONS OF SIN X = K** Plus $n\pi$; the sign flips with n, covering both branches.
$\pm \arccos k$	**ALL SOLUTIONS OF COS X = K** Plus $2n\pi$; both branches at once, since cosine is even.

LAWS OF SINES AND COSINES

A, B, C vs a, b, c	**ANGLES; SIDES** Each lowercase side is opposite its capital angle.
(a, A)	**A SIDE-ANGLE PAIR** One side and the angle across from it.
$s,\ R$	**SEMIPERIMETER; CIRCUMRADIUS** $s = \frac{a+b+c}{2}$; R the radius of the circle through the vertices.

SAME MARKS, DIFFERENT JOB

$\sin^2 x$ vs $\sin^{-1} x$	A power of the value against the inverse function.
$\sin 2$	Two radians, not two degrees.
A	A vertex angle in a triangle; the amplitude on a graph.
$\pm$	Both branches of the solution, not a choice of one.